# Lab 8: Model Comparison and Regularization (pre-lab lecture)

Instructor notes. Part 1: prep. Part 2: podium and circulating cheat sheet

## Overview (instructor only)

| Beat | Mode | Time |
|---|---|---|
| 0. Stan install check (before the lecture) | laptops | ~10 min |
| 1. Big picture | BOARD | ~3 min |
| 2. Catch-up from Monday (**update after Monday**) | BOARD → R | ~12 min |
| 3. Penalties as budgets: defining $t$ | BOARD | ~8 min |
| 4. Tools for the lab | BOARD | ~5 min |
| 5. Cross-validation | BOARD | ~7 min |
| 6. Launch Act 1 | BOARD | ~2 min |

- Lecture ~37 min; with the Stan check, Acts start ~48 min in, leaving just under 2 h. Act 2 is take-home capable.
- **Don't give away:** the 2-D budget picture (Act 1 Part 2), whether to penalize $\beta_0$ or why standardize (Act 2 1a-1b), or step-by-step CV pseudocode (Act 2 1c).

## Part 1: Lecture content

### 0. Stan check (~10 min, before lecture)

- Everyone runs the myxo model from the Week 9 take-home WOD. Pair working laptops with broken ones; collect unresolved cases for office hours rather than holding the room.

### 1. Big picture (~3 min) [BOARD]

- Still COMPARE (2): which features earn a place in the model?
- Today's data: 60 ponds, 8 pond features; Lab 7's model with more predictors. Temperature and elevation: $r$ = -0.94.

## 2. Catch-up from Monday (~12 min) [BOARD → R]

- **Placeholder; update after Monday.** Most likely: Monday section 5 (ridge and lasso on the creepers: 1-D penalty sketch, glmnet paths, `cv.glmnet`, the shrinkage pause). Run it from the Monday section of `CODE_Week10_ModelComparison.R`.
- If Monday finished: use the time for a longer section 5 here, or start the Acts early.

## 3. Penalties as budgets: defining $t$ (~8 min) [BOARD]

**Definition (board).**

1. The ridge fit for a given $\lambda$:

$$\hat{\beta}_\lambda = \arg\min_\beta \left[\text{NLL}(\beta) + \lambda \sum_j \beta_j^2\right]$$

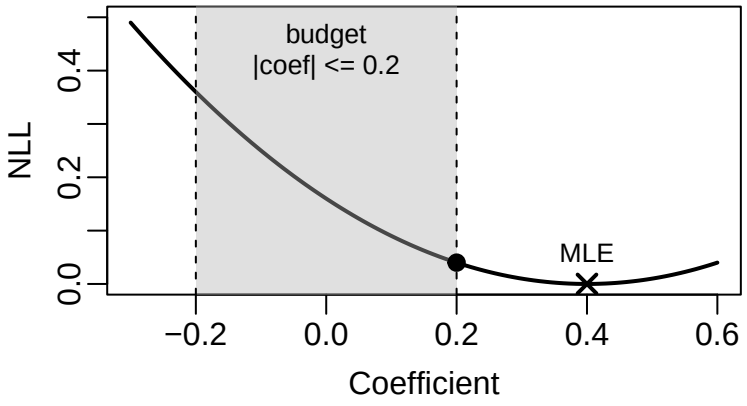
2. Define the budget as the size of that answer:

$$t^2 = \sum_j \hat{\beta}_{\lambda,j}^2 \quad \text{(ridge)}, \qquad t = \sum_j |\hat{\beta}_{\lambda,j}| \quad \text{(lasso)}$$

3. Claim: $\hat{\beta}_\lambda$ is also the **lowest-NLL point inside the budget** $\sum_j \beta_j^2 \le t^2$.
4. Why (one line): if some $\beta^*$ inside the budget had a lower NLL, its penalty $\lambda \sum \beta_j^{*2} \le \lambda t^2$ would be no bigger than $\hat{\beta}_\lambda$'s, so its penalized total would be lower too. But $\hat{\beta}_\lambda$ has the lowest total. Contradiction. (Same argument for the lasso with $|\beta|$.)
5. So each $\lambda$ sets a $t$: $\lambda = 0$ gives $t =$ size of the MLE (the budget doesn't bind); $\lambda \to \infty$ gives $t \to 0$.

**Worked example (one coefficient; the figure).** $\text{NLL}(\beta) = (\beta - 0.4)^2$, so the MLE is 0.4.

- Ridge: $\frac{d}{d\beta}\left[(\beta - 0.4)^2 + \lambda\beta^2\right] = 2(\beta - 0.4) + 2\lambda\beta = 0 \Rightarrow \hat{\beta} = 0.4/(1 + \lambda)$. With $\lambda = 1$: $\hat{\beta} = 0.2$, so $t = 0.2$
- Lasso ($\beta > 0$): $2(\beta - 0.4) + \lambda = 0 \Rightarrow \hat{\beta} = 0.4 - \lambda/2$. With $\lambda = 0.4$: $\hat{\beta} = 0.2$, so $t = 0.2$. Once $\lambda \ge 0.8$, $\hat{\beta} = 0$ (the corner)
- Figure: inside the budget $|\beta| \le 0.2$, the lowest NLL is at the budget's edge, $\beta = 0.2$

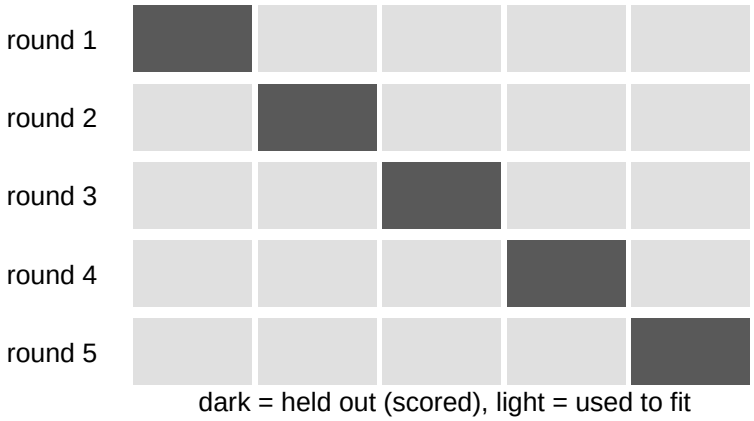


- **Sticking point:** $t$ isn't chosen separately; $\lambda$ (with the data) sets it. We tune $\lambda$ by CV; $t$ is a way to picture what $\lambda$ does
- Stop at one coefficient; Act 1 does two

## 4. Tools for the lab (~5 min) [BOARD]

- `glmnet(X, y, family = "binomial", alpha = )`: `alpha = 0` ridge, `1` lasso
- Binomial counts: `y = cbind(failures, successes)` (the reverse of `glm()`'s order)
- **Sticking point:** `glmnet` scales the NLL by the sample size, so its $\lambda$ is not the $\lambda$ in a hand-coded penalty. Compare coefficients, not $\lambda$ values.

## 5. Cross-validation (~7 min) [BOARD]




- Fit to 4/5 of the ponds, score the held-out 1/5 by its NLL, rotate
- Choose the $\lambda$ with the best held-out score; `lambda.1se` = simplest within 1 SE
- Callback: Monday's WOD "new plot" made systematic
- Folds are random: set a seed to reproduce

## 6. Launch Act 1 (~2 min)

- Laptops closed; Act 2 can go home.

# PART 2: PODIUM AND CIRCULATING CHEAT SHEET

*Tight on purpose. Detail: Part 1 and the two keys.*

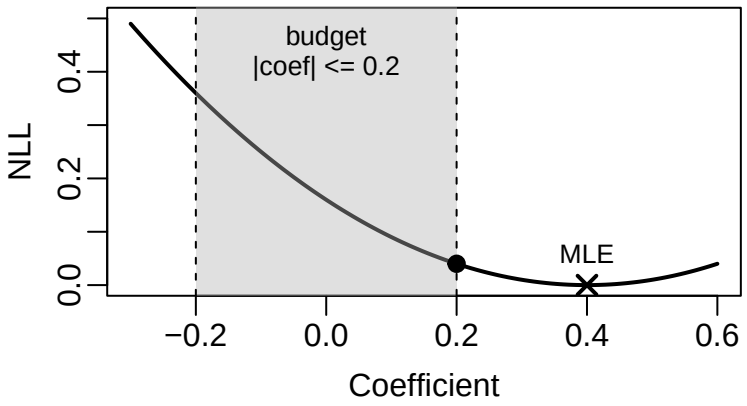
**Order:** Stan check (~10 min), then lecture (~37 min), then Act 1. **Don't give away:** the 2-D budget picture (Act 1), whether to penalize $\beta_0$ or why standardize (Act 2 1a-1b), or step-by-step CV (Act 2 1c).

### 1-2. Big picture; catch-up (~15 min)

- Still COMPARE (2): which pond features earn a place in the model?
- Today's data: 60 ponds, 8 features; Lab 7's logistic model with more predictors. Temperature and elevation are tangled ($r$ = -0.94)
- Catch-up from Monday: **update after Monday**

## 3. Defining the budget $t$ (~8 min) [BOARD]

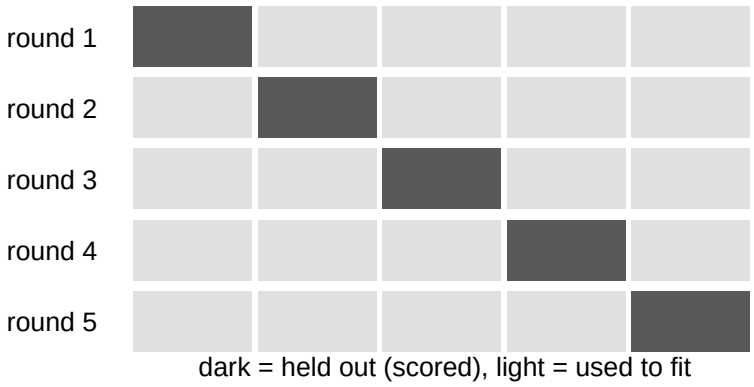
- Ridge fit: $\hat{\beta}_\lambda$ minimizes $\text{NLL}(\beta) + \lambda \sum \beta_j^2$
- Define $t$ as the size of that answer: $t^2 = \sum \hat{\beta}_{\lambda,j}^2$ (lasso: $t = \sum |\hat{\beta}_{\lambda,j}|$)
- Claim: $\hat{\beta}_\lambda$ is the lowest-NLL point with $\sum \beta_j^2 \leq t^2$
- Why: a point inside the budget with lower NLL has penalty $\leq \lambda t^2$, so a lower penalized total too. Contradiction
- $\lambda = 0$: $t$ = size of the MLE. $\lambda \to \infty$: $t \to 0$




- Example, $\text{NLL} = (\beta - 0.4)^2$: ridge $\hat{\beta} = 0.4/(1+\lambda)$, so $\lambda = 1$ gives $t = 0.2$. Lasso $\hat{\beta} = 0.4 - \lambda/2$, so $\lambda = 0.4$ gives $t = 0.2$; $\lambda \geq 0.8$ gives 0
- We tune $\lambda$ (by CV); $t$ just pictures it. Stop at one coefficient; Act 1 does two

## 4-5. Tools; cross-validation (~12 min) [BOARD]

- `glmnet(X, y, family = "binomial", alpha = )`: `alpha = 0` ridge, `1` lasso
- Binomial counts: `y = cbind(failures, successes)`, the reverse of `glm()`'s order
- **Sticking point:** glmnet scales the NLL by sample size, so its $\lambda$ isn't your hand-coded $\lambda$. Compare coefficients, not $\lambda$s




- Fit to 4/5 of the ponds, score the held-out 1/5 by its NLL, rotate; pick the $\lambda$ with the best total
- `lambda.1se` = simplest model within 1 SE of the best. Folds are random: set a seed
- Callback: Monday's WOD "new plot," made systematic

## Act-by-Act quick reference

### Act 1 (paper)

- Best by AIC: Model 5 (temp + bullfrogs + canopy). Model 6 adds elevation and sits within 2, but its NLL drops only 0.38: an uninformative parameter
- LRTs: 5 vs. 6, $D = 0.76$ (1 df, cutoff 3.84); 5 vs. 7, $D = 3.98$ (5 df, cutoff 11.07). Keep Model 5 both times
- Temperature vs. elevation: these data can't separate them; need ponds where they come apart, or an experiment
- Part 2 (temperature vs. canopy, budget $t = 0.3$): ridge $\approx$ (-0.26, 0.14); lasso (-0.30, 0.00), on the corner, so **canopy is exactly 0** even though canopy truly matters
- 2b: ridge essentially never zeros a coefficient (a circle has no corners). 2c: the lasso zeros one when the lowest contour hits a corner first: small budgets, one effect weak relative to the other. Correlated pairs (contours parallel to an edge) tend to keep both, which sets up Act 2 3b

### Act 2 (coding)

- 1a: don't penalize $\beta_0$ (it sets overall prevalence). 1b: standardize so one penalty treats every coefficient alike
- Check values: temperature -0.64 at $\lambda = 0$, -0.41 at $\lambda = 10$; elevation 0.16 $\rightarrow$ 0.30 (ridge pulls the correlated pair together)
- Lasso (`set.seed(1)`, 10 folds) keeps temp, elev, canopy, bullfrogs
- Common bugs: `cbind()` order in glmnet; penalizing $\beta_0$; forgetting to standardize; `optim()`'s default Nelder-Mead stalling with 9 parameters (use BFGS)